



First demonstration of a 400 Gb/s 4 λ CWDM TOSA for datacenter optical interconnects

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Abstract: We present the first demonstration of a 4 λ transmitter optical sub-assembly (TOSA) on the coarse wavelength division multiplexing (CWDM) grid, i.e., 20 nm spacing, targeting 400G-FR4 requirements over 2 km. The TOSA is based on uncooled InP external modulated laser (EML) technology and it utilizes four EMLs followed by a CWDM multiplexer. We characterize the performance of the TOSA versus received optical modulation amplitude (OMA), number of equalizer taps, reach, modulation format, TOSA case temperature, and bit rate. Four 53 Gbaud 4-level pulse amplitude modulation (PAM4) RF signals are used to drive the TOSA achieving a net rate of 400 Gb/s. Results reveal that 400 Gb/s can be transmitted over 2 km of single mode fiber (SMF) at a bit error rate (BER) below the KP4- forward error correction (KP4-FEC) threshold (i.e., 2.4×10^{-4}) using only a 5 tap feed forward equalizer at the receiver. To the best of our knowledge, this is the first demonstration of 400 Gb/s using a 4 λ CWDM TOSA over 2 km of SMF. Moreover, we achieve 400 Gb/s and 600 Gb/s over 20 km and 10 km below KP4-FEC and the 7% hard-decision FEC (HD-FEC) (i.e., 3.8×10^{-3}) thresholds, respectively, without optical amplification. Furthermore, we show the performance of the TOSA against temperature, where it shows no significant change in the BER performance from 20 °C to 60 °C. Finally, we compare the performance of PAM2, PAM4, and PAM8 modulation formats where we show the possibility of achieving 400 Gb/s aggregate bit rate using 42 Gbaud PAM8 modulation format at the expense of utilizing a stronger FEC.

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1. Introduction

The continuous increase in datacenter traffic drives the development of high speed optical transceivers for datacenter optical interconnects [1]. To cope with such increases, 100 Gb/s transceivers based on 4×25 Gb/s non-return to zero (NRZ) are currently being deployed. The next generation of Ethernet optical transceivers will operate at 400 Gb/s. Scaling from 100 Gb/s to 400 Gb/s requires the increase of single lane bit rate and/or number of lanes. Several ≥ 100 Gb/s single carrier results have been recently reported using pulse amplitude modulation (PAM), dual polarization PAM, discrete multi-tone (DMT), and multi-band carrierless amplitude phase modulation (multi-CAP) [2–7]. Also, few ≥ 400 Gb/s wavelength-division multiplexing (WDM) based demonstrations have recently been reported using discrete multi-tone (DMT) [8] and half-cycle 16QAM Nyquist-subcarrier multiplexing [9].

In 2017, the IEEE standardized the 400GBase Ethernet specifications [10]. The standard over 2 and 10 km of single mode fiber (SMF) is: 25 Gbaud, 4-level pulse amplitude modulation (PAM4), and eight WDM lanes with 800 GHz spacing in the O-band [10]. In 2018, the 100G Lambda Multi-Source Agreement (MSA), supported by a broad industry consortium, has been recently formed providing specifications for 100 Gb/s per lane [11]. Leveraging the 100 Gb/s per lane, the MSA established the 400 Gb/s transceivers technical specifications over 2 km of SMF (400G-FR4): 50 Gbaud PAM4, and four lanes on the coarse WDM (CWDM) grid, i.e., 20 nm spacing [11].

Table 1 summarizes 400 Gb/s demonstrations compared to this work, including [12] which represents a recently published demonstration that is compliant with the IEEE 400GBASE-LR8 specifications. In [13], the first real-time transmission of 400 Gb/s (8×50 Gb/s) PAM-4 signals for data center interconnects up to 100 km of SMF is successfully demonstrated using discrete components and bulk modulation. In [14], eight-channel hybrid multi-chip module comprising InP lasers, silicon photonic modulators, and parallel SMFs, all connected via photonic wire bonds is presented which can achieve 400 Gb/s aggregate bit rate. However, each lane is tested individually using RF probes while cross-talk between the modulators is expected to degrade the performance when operating simultaneously [15]. Also, the module is far from commercial

Table 1. Comparison of our work with previous 400 Gb/s demonstrations.

Ref.	Tx structure	Package	Band	No. of channels	Spacing	Reach	Optical Amp.	FEC threshold
[12; 2017]	8 DMLs + Mux	CFP8	O-band	8 λ	800 GHz (LWDM)	10 km	No	2.4×10^{-4}
[12; 2017]	8 EMLs + Mux	CFP8	O-band	8 λ	800 GHz (LWDM)	10 km	No	2.4×10^{-4}
[13; 2016]	2 MZMs (Bulk mod.)	discrete	C-band	8 λ	50 GHz (DWDM)	100 km	Yes	5.2×10^{-5}
[14; 2017]	8 DFBs + 8 SiP MZMs	chip-level	C-band	8 fibers	PSM*	2 km	No	3.8×10^{-3}
[16; 2015]	4 EMLs	TOSA	O-band	4 λ	800 GHz (LWDM)	2 km	No	3.8×10^{-3}
This work	4 EMLs + MUX	TOSA	O-band	4λ	20 nm (CWDM)	20 km	No	2.4×10^{-4}

* Parallel single mode fibers.

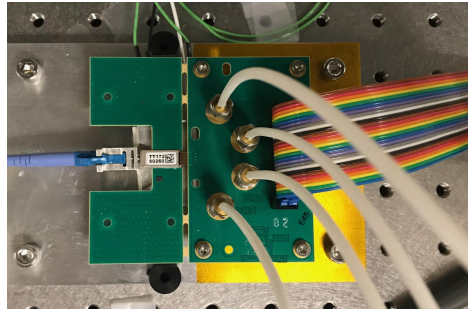
products due to the operating wavelength, performance, size, reliability, and yield. In [16], 465 Gb/s net rate has been achieved using four commercial 25 Gb/s external modulated lasers (EMLs) on the LAN-WDM (LWDM) grid, i.e., 800 GHz spacing. However, equalization, post filtering, and maximum likelihood sequence estimation are needed to compensate for the induced inter-symbol interference due to the EML limited bandwidth.

In this paper, we present to the best of our knowledge the first transmission demonstration of a four lane 400 Gb/s CWDM transmitter optical sub-assembly (TOSA) that meets key 400G-FR4 MSA specifications and can support a QSFP-DD transceiver form factor. Four lanes, each running at 53 Gbaud PAM4, are used to drive the four lane CWDM-TOSA with a net data rate of 400 Gb/s. We report the performance of the TOSA versus received optical modulation amplitude (OMA), number of taps, reach, modulation format, TOSA case temperature, and bit rate. Results reveal that using only a 5-tap T-spaced feed forward equalizer (FFE) at the receiver, 400 Gb/s transmission over 2 km of SMF is achieved at a bit error rate (BER) below the KP4 forward error correction (KP4-FEC) threshold of 2.4×10^{-4} at -5 dBm OMA for the worst lane. In addition, we demonstrate 400 Gb/s over up to 20 km reach below the KP4-FEC without optical amplification. Moreover, we report 600 Gb/s over 10 km of SMF at aBER below the hard decision FEC (HD-FEC) of 3.8×10^{-3} using the same CWDM-TOSA enabled by an 11-tap FFE at the receiver. Furthermore, we show that reach can be extended to 40 km using an optical amplifier for the lanes where the dispersion effect is not dominant. The TOSA performance versus operating temperature is also assessed and we show that the TOSA can operate over temperature range from 20°C to 60°C without performance degradation. Finally, we compare the performance of PAM2, PAM4, and PAM8 modulation formats, where we show that 100 Gb/s per lane can be achieved using the same TOSA running at 35 Gbaud PAM8.

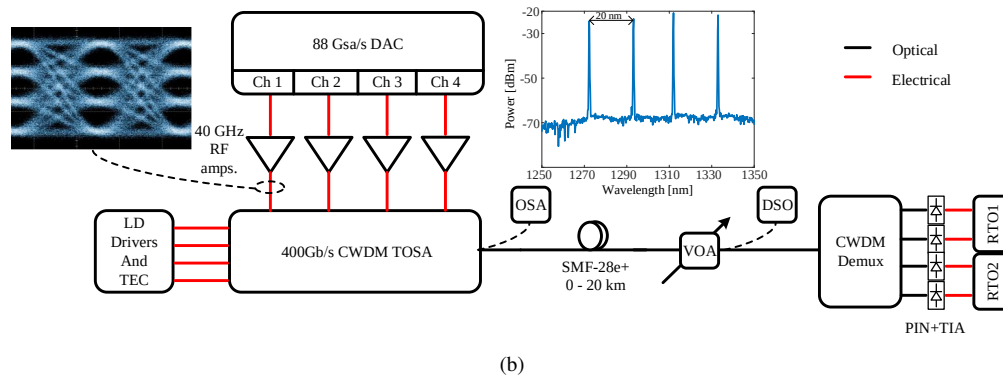
The rest of this paper is organized as follows. In section 2, the device details are explained, and the experimental setup is introduced. In section 3, the system-level experimental results for the CWDM-TOSA are presented. Finally, the paper is concluded in section 4.

2. Experimental setup

The TOSA is a 400 Gb/s CWDM device based on uncooled InP EML technology and it utilizes four EMLs followed by a CWDM multiplexer. The nominal wavelengths of the EMLs are 1271 nm, 1291 nm, 1311 nm, and 1331 nm. The compact size of the TOSA can support QSFP-DD transceiver form factor. As shown in the photo in Fig. 1(a), the TOSA was soldered to an RF board with 2.4 mm RF connectors for the input RF signals, and DC connections for the laser current sources. Also, a temperature controller (TEC) is needed to sweep the TOSA case temperature as shown later in Fig. 6(b). Figure 1(b) shows the experimental setup. Four lanes of an ILX-3916 laser diode controller are used to drive the lasers. An 8-bit digital-to-analog converter (DAC)



(a)



(b)

Fig. 1. (a) An image for the TOSA soldered to the RF board in the test bed, and (b) experimental setup used for the 400G CWDM-TOSA testing. Insets: 53 Gbaud RF signal out of the amplifier, and optical spectrum out of the TOSA. DAC: digital-to-analog converter, TEC: temperature controller, OSA: optical spectrum analyzer, VOA: variable optical attenuator, DSO: digital sampling oscilloscope, and RTO: real time oscilloscope.

running at 88 GSa/s is used to generate four PAM4 signals. The output RF signals from the four lanes of the DAC are amplified using four 40 GHz RF amplifiers followed by four bias-Ts each with a 3-dB bandwidth of 65 GHz for applying the modulator bias. A 53 Gbaud PAM4 eye diagram after the bias-T for one of the driving lanes is shown in the top-left inset of Fig. 1(b). At the transmitter side, we only pre-compensate the response of the DAC and RF amplifier. The optical spectrum of the 400 Gb/s CWDM signal at the output of the TOSA captured using an optical spectrum analyzer (OSA) is shown in the top-right inset in Fig. 1(b). The optical signal is then launched into various lengths of SMF (Corning SMF-28e+) covering reaches ranging from 500 m to 20 km without optical amplification. A variable optical attenuator (VOA) is added before the receiver to sweep the received signal power, and consequently the received OMA. At the receiver side, a commercial CWDM demultiplexer followed by four 40 GHz photoreceivers (PD+TIA) are added. The four signals out of the photoreceivers are sampled at 160 GSa/s by two 62 GHz real time oscilloscopes (RTOs) and stored for offline processing. The offline processing includes: resampling, equalization, and bit error counting. In order to minimize the required digital signal processing (DSP), we use only 5 tap T-spaced FFE for the 400 Gb/s results which compensates for the combined response of the RF board, TOSA, and the photoreceiver (per above, the response of the DAC and RF amplifier has already been pre-compensated).

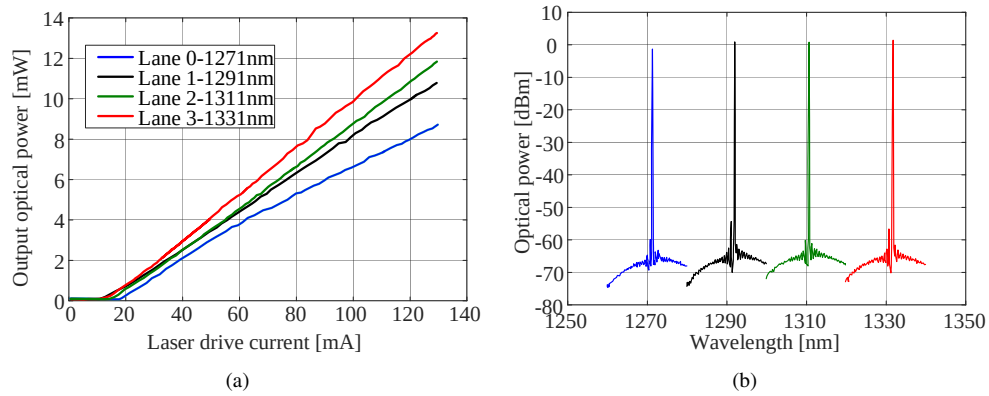


Fig. 2. (a) Light-current characteristics, and (b) measured optical spectra for the four CWDM lasers.

3. Experimental results

Figure 2(a) presents the light-current characteristics for the four CWDM lasers at 40 °C. The value of the threshold current for the four lasers ranges from 11.2 to 17.5 mA. The measured optical spectra are shown in Fig. 2(b), where the side mode suppression ratio (SMSR) is more than 55 dB for the four lasers.

Figure 3 shows clear open optical eye diagrams captured using a 65 GHz optical sampling head of a Keysight digital sampling oscilloscope (DSO) in the back-to-back (B2B) and 2 km cases for all lanes after the demultiplexer. A 5-tap T-spaced FFE is applied to the eye diagrams using the built-in FFE in the DSO.

Figure 4(a) presents the B2B BER performance versus OMA for all lanes running at 53 Gbaud PAM4 yielding a net rate of 400 Gb/s. It can be observed that all four lanes can achieve a BER below the KP4-FEC threshold at an OMA of approximately -5 dBm for the worst lane. Lane 2 shows a slightly worse performance compared to the other lanes, which can be attributed to a slightly smaller bandwidth. Similarly, we show in Fig. 4(b) the BER performance versus received OMA when the symbol rate is increased to 64 Gbaud while having the same number of taps.

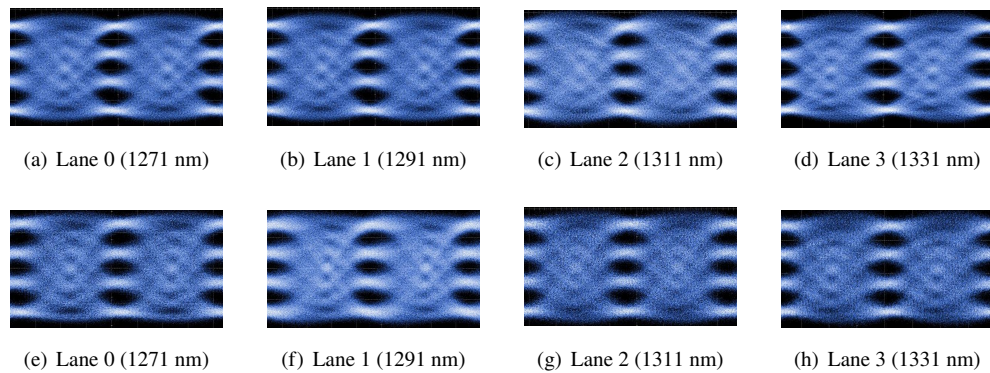


Fig. 3. (a)-(h) Optical eye diagrams for the four received lanes at the demultiplexer output equalized using a 5 tap FFE on the digital sampling oscilloscope in the B2B (top) and 2 km (bottom) cases.

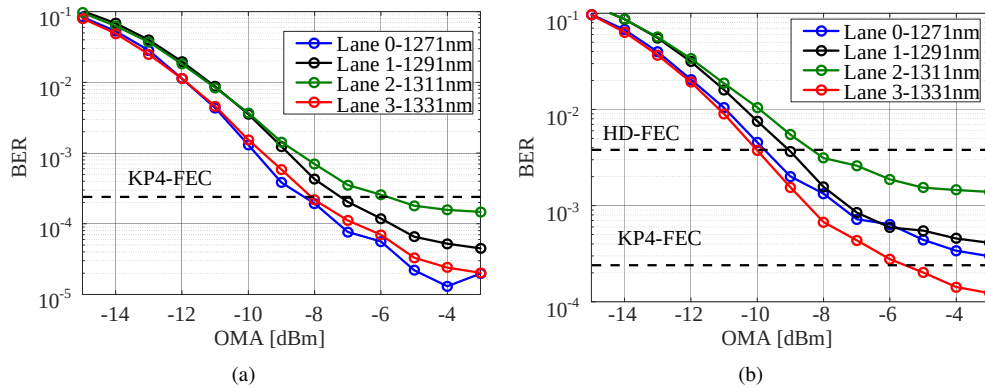


Fig. 4. (a) BER performance versus received OMA for the four lanes running at 53 Gbaud each in the B2B case equalized using a 5 tap FFE, and (b) BER performance versus received OMA for the four lanes running at 64 Gbaud each in the B2B case equalized using a 5 tap FFE.

In Fig. 5(a), the transmission performance of all lanes over SMF for 400 Gb/s net rate is tested over reaches ranging from 500 m to 20 km without optical amplification. The OMA is fixed at -4 dBm for all points, and the receiver equalizer taps are fixed at 5 taps. It can be observed that 400 Gb/s can be transmitted over as much as 20 km of SMF without optical amplification at a BER below the KP4-FEC threshold. To the best of our knowledge, this is the first experimental demonstration of 400 Gb/s transmission over 20 km of SMF using a four lane CWDM TOSA with all lanes simultaneously modulated. For lane 0 (1271 nm), we see a larger penalty compared to the other lanes after 20 km transmission, which is expected since this lane is more than 40 nm away from the SMF zero dispersion wavelength. However, we can still achieve 20 km transmission on this channel using a 5 tap equalizer, and increasing the equalizer taps can significantly improve the performance. Figure 5(b) presents the BER performance for all lanes driven by 75 Gbaud PAM4 signals, yielding 600 Gb/s bit rate (75 Gbaud/lane) over 10 km reach below the HD-FEC threshold (3.8×10^{-3}) using an 11 tap T-spaced FFE. Figure 5(c) presents the performance versus the number of FFE taps for lane 0, which is the lane most affected by fiber dispersion, at 75 Gbaud PAM4 in the B2B and 10 km cases. Increasing the number of taps improves the BER performance until reaching 31 taps where further increases show a small BER improvement. Next, we sweep the bit rate for lane 0 at different reaches using 5 and 31 taps in Fig. 5(d). It can be observed that up to 70 Gbaud over 10 km reach can be received below the HD-FEC threshold using only 5 taps at the receiver. Increasing the number of taps to 31 taps, 80 Gbaud over 2 km reach can be received at a BER below the HD-FEC threshold.

Next, we extend the reach to 40 km by adding a praseodymium doped fiber amplifier as a pre-amplifier. Figure 6(a) show the BER performance versus bit rate over 40 km reach for different lanes. The number of taps is increased to 31 FFE taps to compensate for the dispersion over 40 km except lane 0 where the dispersion is more pronounced and more aggressive equalizer is needed (e.g., using maximum likelihood sequence detection). Lane 2 shows the best BER performance where 128 Gb/s can be received after 40 km at a BER below the KP4-FEC since its wavelength is the closest to the fiber's zero dispersion wavelength. Also, the BER performance degrades for lanes 1 and 3 where 128 Gb/s and 90 Gb/s can be received at a BER below the HD-FEC, respectively. Furthermore, we assess the effect of the TOSA case temperature on the BER performance for all lanes running at 53 Gbaud over 10 km reach in Fig. 6(b). The temperature is varied from 20 °C up to 60 °C, where the received power is kept constant using the VOA since the laser power decreases with increasing temperature. It can be observed that

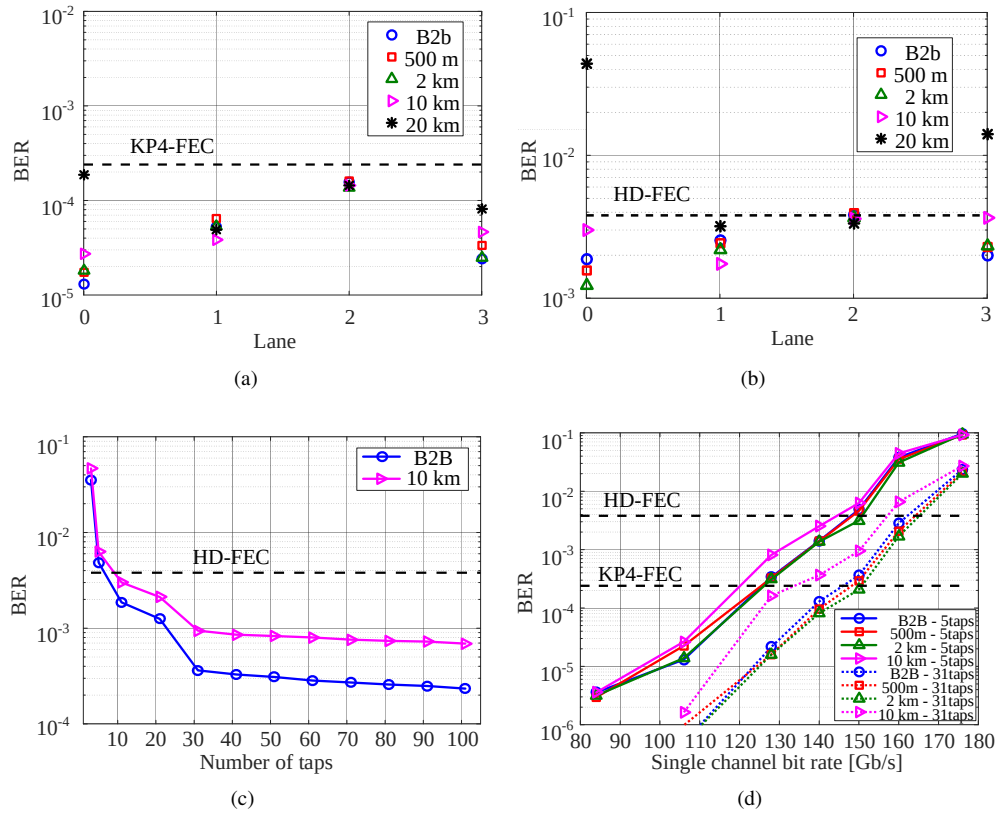


Fig. 5. (a) BER performance for all lanes running at 400 Gb/s aggregate net rate over different reaches using 5 tap receiver FFE, (b) 75 Gbaud per lane (600 Gb/s) BER performance over different reaches using 11 tap FFE, (c) BER performance versus number of receiver FFE taps for lane 0 running at 75 Gbaud in the B2B and 10 km cases, and (d) BER versus bit rate using 5 and 31 receiver FFE taps at -5 dBm received signal power for lane 0.

varying the temperature over this range has a negligible effect on the BER performance, and 400 Gb/s transmission over 10 km can be achieved over the entire temperature range.

Finally, we compare the BER performance versus symbol rate for PAM2, PAM4, and PAM8 modulation formats for different reaches for lane 0 in Fig. 7(a). Also, we add the BER performance versus bit rate for more clarity in Fig. 7(b). To reiterate here, the modulator transfer function is not precompensated at the transmitter side. The number of receiver FFE taps is fixed at 5, 5, and 51 taps for PAM2, PAM4, and PAM8, respectively. Eye diagrams after receiver DSP for 106 and 105 Gb/s signals after 10 km of SMF using PAM4 and PAM8 modulation formats, respectively, are shown in Fig. 7(c). The received signal power was kept at -5 dBm for all curves. For the PAM2 signal, a bit rate up to 88 Gb/s over as much as 10 km can be transmitted at a BER below the HD-FEC threshold. It is interesting to notice that to achieve 100 Gb/s per lane while still having low complexity, PAM4 modulation format is favored over PAM2. However, a higher BER is achieved and a relatively stronger FEC is needed, i.e., KP4-FEC, comparing 50 Gbaud PAM2 and PAM4. Similarly, we can see a crossing point above 150 Gb/s between PAM4 and PAM8 modulation formats which suggests that pushing further will require using PAM8 instead to decrease the bandwidth requirement compared to PAM4. What we forecast in our paper that with the development of better electrical signal generators compared to the limited effective number

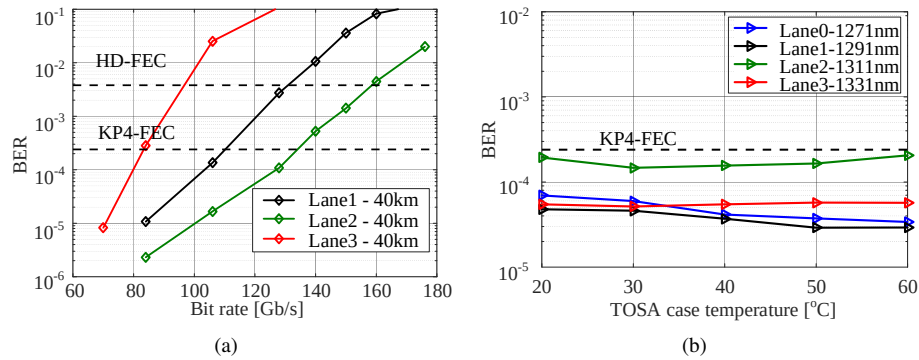


Fig. 6. (a) BER versus bit rate after 40 km reach using 31 FFE taps, and (b) BER versus TOSA case temperature at constant received signal power for 400 Gb/s net rate over 10 km reach.

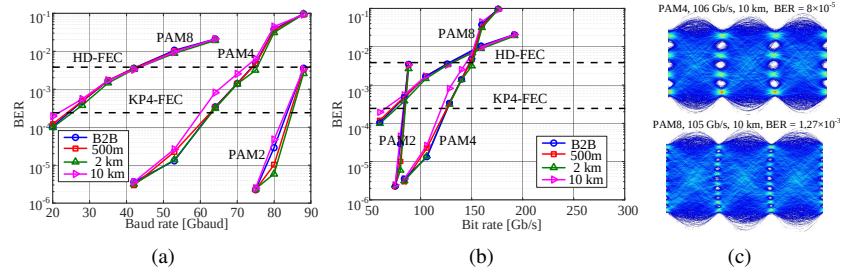


Fig. 7. BER versus (a) baud rate and (b) bit rate for different modulation formats over different reaches, and (c) eye diagrams after receiver DSP for PAM4 and PAM8 modulation formats running at 53 and 35 Gbaud, respectively.

of bits (ENOB) and bandwidth of our DAC, PAM8 signal generation can be less complex. Also, better modulators with more superior linearity are needed for the PAM8 electrical-to-optical mapping. Hence, operating at lower baud rate can achieve equal and higher net bit rates compared to PAM4 modulation formats at the expense of adopting a stronger FEC.

4. Conclusion

We present the first demonstration of 400 Gb/s (4×100 Gb/s) CWDM-TOSA targeting 400G-FR4 requirements. We present a detailed system-level study of the CWDM-TOSA versus several parameters. Four 53 Gbaud PAM4 RF signals are used to drive the TOSA achieving a net rate of 400 Gb/s. Results reveal that 400 Gb/s can be transmitted over up to 2 km of SMF at a BER below the KP4-FEC threshold using only a 5 tap FFE at the receiver at -5 dBm OMA. To the best of our knowledge, this is the first demonstration of 400 Gb/s using a 4λ CWDM TOSA over 2 km of SMF. Also, we show the TOSA is capable of achieving 400 Gb/s over 20 km, and 600 Gb/s over 10 km below the KP4- and HD-FEC thresholds, respectively, without optical amplification. Furthermore, we show the performance of the TOSA against case temperature, where it shows no significant change in the BER performance from 20°C to 60°C . Finally, we compare the performance of PAM2, PAM4, and PAM8 modulation formats where we show the possibility of achieving a higher bit rate using PAM8 modulation format at the expense of utilizing a stronger FEC.